



# Datasheet

## ESP32 MiniPLC

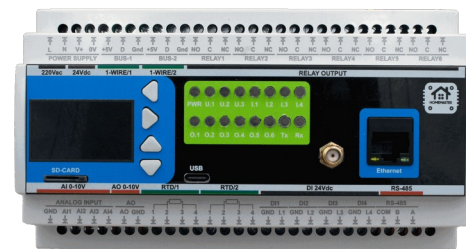
Part No.: MiniPLC-R1

Manufacturer: ISYSTEMS AUTOMATION S.R.L.

Hardware Version: V1.0

Architecture: Modular (MCU Board + RelayBoard + USB Board)

Controller: ESP32-WROOM-32U



## Functional Overview

The HomeMaster MiniPLC is a professional, open-source DIN rail controller built around the ESP32 platform, designed for robust and scalable smart automation in residential, commercial, and light industrial environments. It combines extensive onboard I/O—including relays, digital and analog inputs, temperature sensors, and a user interface—with native Home Assistant integration via pre-installed ESPHome.

Engineered for reliability and flexibility, the MiniPLC supports multiple power input options (24V DC or wide-range AC/DC) and features a non-isolated RS-485 (MAX485; transceiver shares logic ground) Modbus interface for expansion with a wide range of compatible I/O modules. Its local processing capability ensures continued operation even when network or cloud connectivity is lost, making it suitable for mission-critical applications.

This controller operates as a complete standalone automation system using its comprehensive onboard I/O, while also offering seamless expansion via the RS-485 bus to connect HomeMaster smart modules for energy metering, lighting control, security, and more. It provides a versatile, ready-to-integrate platform for HVAC control, energy management, security systems, lighting automation, and custom industrial applications.

The system is implemented using three hardware boards: MCU board, USB board and Relay board.

| Board       | Function                               |
|-------------|----------------------------------------|
| MCU Board   | CPU, analog acquisition, Ethernet, USB |
| Relay Board | Power supply, relays, digital inputs   |
| USB Board   | USB-C and SD storage                   |

## Technical Specifications

| Specification      | Details                                                                                                                                                       |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Microcontroller    | ESP32-WROOM-32U-N16 (dual-core)                                                                                                                               |
| Storage            | MicroSD card (SPI interface, power-switched 3.3 V rail)                                                                                                       |
| Power Input        | 24 V DC nominal or 85–265 V AC, 47–63 Hz (L / N)                                                                                                              |
| RTC                | RTC (PCF8563); backup battery holder, battery not fitted                                                                                                      |
| Document revision  | DS-MiniPLC Rev. B · 2026-09 · Hardware V1.0                                                                                                                   |
| Digital Inputs     | 4 × isolated 24 V DC discrete inputs (DIx + GNDx), supporting dry-contact closure protected by per-channel PTC fuse, TVS surge suppression, and EMI filtering |
| Relay outputs      | 6 × SPDT (NO / C / NC); 3 A @ 250 VAC per contact (module limit)                                                                                              |
| Analog Inputs      | 4 × 0–10 V, 16-bit (ADS1115)                                                                                                                                  |
| Analog Outputs     | 1 × 0–10 V, 12-bit (MCP4725)                                                                                                                                  |
| Temperature Inputs | 2 × RTD (PT100/PT1000 via MAX31865), 2 × 1-Wire (DS18B20 compatible)                                                                                          |
| Display            | 128 × 64 OLED (SH1106)                                                                                                                                        |
| User Interface     | 4 buttons, 16 LEDs: PWR ×1, user ×2 (U.1, U.2), status ×1 (U.3), RX/TX ×2, DI ×4, relay ×6                                                                    |
| RS-485             | non-isolated RS-485 (MAX485; transceiver shares logic ground); half-duplex Modbus RTU, not galvanically isolated                                              |
| Wi-Fi              | Wi-Fi (ESP32)                                                                                                                                                 |
| Ethernet           | Optional Ethernet (LAN8720 PHY)                                                                                                                               |
| USB                | USB-C (ESD protected, CC detection, data to ESP32)                                                                                                            |

### Top Terminal Block (1–28)

| Pos | Marking | Signal   | Description           |
|-----|---------|----------|-----------------------|
| 1   | L       | POWER_AC | AC line (MAINS)       |
| 2   | N       | POWER_AC | AC neutral (MAINS)    |
| 3   | V+      | POWER_DC | 24 V DC input         |
| 4   | 0V      | POWER_DC | 24 V DC return        |
| 5   | +5V     | BUS1     | 1-Wire bus 1 supply   |
| 6   | D       | BUS1     | 1-Wire bus 1 data     |
| 7   | Gnd     | BUS1     | 1-Wire bus 1 ground   |
| 8   | +5V     | BUS2     | 1-Wire bus 2 supply   |
| 9   | D       | BUS2     | 1-Wire bus 2 data     |
| 10  | Gnd     | BUS2     | 1-Wire bus 2 ground   |
| 11  | NO      | RELAY1   | Relay 1 normally open |
| 12  | C       | RELAY1   | Relay 1 common        |

|    |    |        |                         |
|----|----|--------|-------------------------|
| 13 | NC | RELAY1 | Relay 1 normally closed |
| 14 | NO | RELAY2 | Relay 2 normally open   |
| 15 | C  | RELAY2 | Relay 2 common          |
| 16 | NC | RELAY2 | Relay 2 normally closed |
| 17 | NO | RELAY3 | Relay 3 normally open   |
| 18 | C  | RELAY3 | Relay 3 common          |
| 19 | NC | RELAY3 | Relay 3 normally closed |
| 20 | NO | RELAY4 | Relay 4 normally open   |
| 21 | C  | RELAY4 | Relay 4 common          |
| 22 | NC | RELAY4 | Relay 4 normally closed |
| 23 | NO | RELAY5 | Relay 5 normally open   |
| 24 | C  | RELAY5 | Relay 5 common          |
| 25 | NC | RELAY5 | Relay 5 normally closed |
| 26 | NO | RELAY6 | Relay 6 normally open   |
| 27 | C  | RELAY6 | Relay 6 common          |
| 28 | NC | RELAY6 | Relay 6 normally closed |

Bottom Terminal Block (1-26)

| Pos | Marking | Signal        | Description               |
|-----|---------|---------------|---------------------------|
| 1   | GND     | ANALOG_INPUT  | Common return for AI1-AI4 |
| 2   | AI1     | ANALOG_INPUT  | Analog input 1, 0-10 V    |
| 3   | AI2     | ANALOG_INPUT  | Analog input 2, 0-10 V    |
| 4   | AI3     | ANALOG_INPUT  | Analog input 3, 0-10 V    |
| 5   | AI4     | ANALOG_INPUT  | Analog input 4, 0-10 V    |
| 6   | AO      | ANALOG_OUTPUT | Analog output, 0-10 V     |
| 7   | GND     | ANALOG_OUTPUT | Analog output             |

|    |     |       |                         |
|----|-----|-------|-------------------------|
|    |     |       | return                  |
| 8  | 1   | RTD1  | RTD 1 terminal 1        |
| 9  | 2   | RTD1  | RTD 1 terminal 2        |
| 10 | 3   | RTD1  | RTD 1 terminal 3        |
| 11 | 4   | RTD1  | RTD 1 terminal 4        |
| 12 | 1   | RTD2  | RTD 2 terminal 1        |
| 13 | 2   | RTD2  | RTD 2 terminal 2        |
| 14 | 3   | RTD2  | RTD 2 terminal 3        |
| 15 | 4   | RTD2  | RTD 2 terminal 4        |
| 16 | GND | DI1   | DI1 return              |
| 17 | I.1 | DI1   | DI1 input               |
| 18 | GND | DI2   | DI2 return              |
| 19 | I.2 | DI2   | DI2 input               |
| 20 | GND | DI3   | DI3 return              |
| 21 | I.3 | DI3   | DI3 input               |
| 22 | GND | DI4   | DI4 return              |
| 23 | I.4 | DI4   | DI4 input               |
| 24 | COM | RS485 | RS-485 signal reference |
| 25 | B   | RS485 | RS-485 data -           |
| 26 | A   | RS485 | RS-485 data +           |

\*Relay outputs are **not internally fused**. External overcurrent protection must be provided in accordance with applicable standards and installation requirements. Use appropriately rated fuses or circuit breakers and consider external contactors for loads exceeding 3 A or for inductive and high inrush loads.

## Installation, Environmental & Mechanical

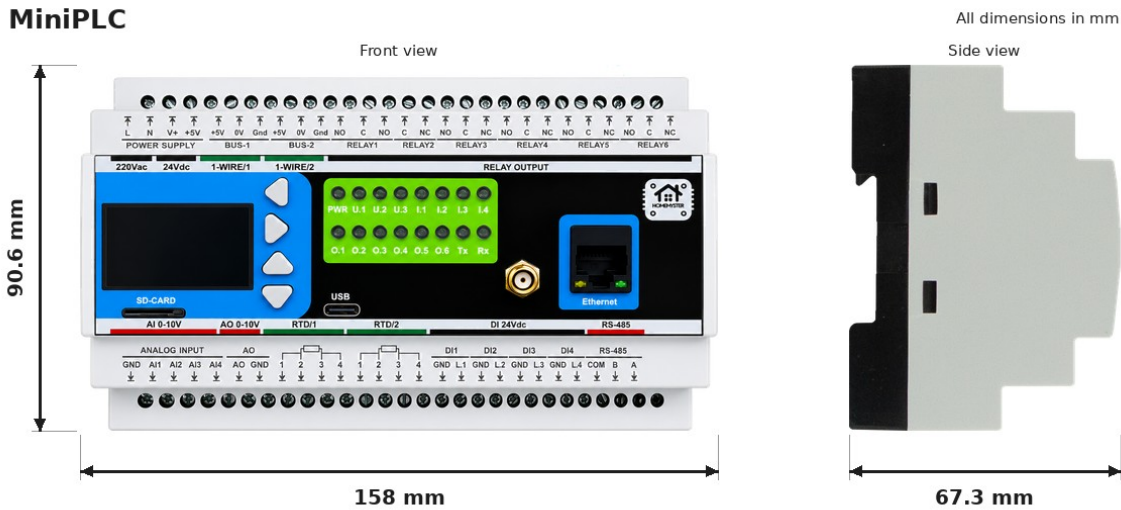
| Category               | Specification         | Details                                                              |
|------------------------|-----------------------|----------------------------------------------------------------------|
| Mechanical & Packaging | Terminal type         | Pluggable screw terminal blocks, 5.08 mm pitch                       |
|                        | Terminal pitch        | 5.08 mm                                                              |
|                        | Wire cross-section    | 0.2–2.5 mm <sup>2</sup> (AWG 24–12)                                  |
|                        | Conductor type        | Solid or stranded copper                                             |
|                        | Stranded wire         | Ferrules recommended                                                 |
|                        | Tightening torque     | 0.4–0.6 Nm                                                           |
|                        | Operating temperature | 0 °C to +40 °C                                                       |
|                        | Storage temperature   | –10 °C to +55 °C                                                     |
|                        | Relative humidity     | 0–90 % RH, non-condensing                                            |
|                        | Ingress protection    | IP20 (inside cabinet only)                                           |
|                        | Installation          | Indoor control cabinet only; not for outdoor or exposed installation |
|                        | Maximum altitude      | 2000 m                                                               |

|  |                  |                                  |
|--|------------------|----------------------------------|
|  | Pollution degree | 2                                |
|  | Dimensions       | 158 × 90.6 × 67.3 mm (L × W × H) |
|  | DIN width        | 9 modules ( ≈ 158 mm)            |
|  | Mounting         | 35 mm DIN rail                   |
|  | Enclosure        | PC/ABS industrial enclosure      |
|  | Net weight       | 300 g                            |
|  | Gross weight     | 450 g                            |
|  | Pack size        | 230 × 140 × 87 mm (L × W × H)    |

Install only inside a control cabinet with ventilation; the cabinet must include a protective front plate covering all module connection terminals and a closing protective door; not for outdoor or exposed installation.

**All wiring terminals must be protected against accidental contact by an insulating front plate, wiring duct, or terminal cover. Exposed live terminals are not permitted.**

### MiniPLC



## Cable Recommendations & Shield Grounding

This section applies to **Analog (0–10V)**, **Temperature (RTD / 1-Wire)**, and **RS-485**. Use shielded, twisted constructions and bond shields correctly to reduce EMI and ground-loop issues.

### *General Routing Rules*

- Route low-level signal cables (Analog/RTD/1-Wire/RS-485) separately from mains, relay outputs, contactors, VFD motor cables, and power wiring.
- If crossing power cables is unavoidable, cross at **90°**.
- Keep cable runs as short as practical and avoid parallel runs with high-current conductors.

### *Analog (0–10V) Cable*

- **Construction:** twisted pair (Signal + GND) per channel
- **Shielding:** overall shield (standard) or individually shielded pairs (high-EMI)
- **Examples:** J-Y(ST)Y (overall shield) or LI2YCY PIMF (shielded twisted pairs; one pair per channel)

### *Temperature Cable*

#### **RTD (PT100/PT1000)**

- **Recommended:** shielded multi-core for 2/3-wire; shielded pairs for best accuracy (4-wire)
- **Examples:** J-Y(ST)Y (overall shield) or LI2YCY PIMF (pairs; e.g. 2×2×0.50 for 4-wire)

#### **1-Wire (DS18B20)**

- **Recommended:** shielded 3-core ( +5V / DATA / GND ) for typical installations
- **High-EMI / long runs:** shielded pairs + overall shield (e.g. LI2YCY PIMF 2×2×0.50)
- **Topology:** daisy-chain (bus). Avoid star wiring.
- **Stubs:** keep sensor stubs  $\leq 0.5$  m.
- **Pull-up (DATA):** 4.7 k $\Omega$  typical; 2.2–3.3 k $\Omega$  for long/heavy loads.

### *RS-485 Cable*

- **Construction:** twisted pair for A/B
- **Characteristic impedance:** 120  $\Omega$  (recommended)
- **Shielding:** overall shield is recommended in cabinets; use individually shielded pairs + overall shield in high-EMI
- **Examples:** J-Y(ST)Y 2×2×0.5 mm<sup>2</sup> or LI2YCY PiMF 2×2×0.50
- Use one twisted pair for A/B. Run COM to every node. Required, not optional: the RS-485 port is not galvanically isolated.

## Shield Grounding

- **Default recommendation:** bond cable shield(s) to cabinet **PE/EMC ground at the PLC end only**.
- **Do not** connect shields to signal terminals (AI/AO/RTD/1-Wire/RS-485 A/B/COM).
- If both ends are in equipotential bonded cabinets, shields may be bonded at both ends using proper 360° clamps.

## Compliance & Certifications

The HomeMaster MiniPLC is CE marked and designed to comply with the applicable European Union directives. The manufacturer, ISYSTEMS AUTOMATION (HomeMaster brand), maintains the technical documentation and a signed EU Declaration of Conformity (DoC).

### EU Directives

EMC 2014/30/EU · LVD 2014/35/EU · RED 2014/53/EU · RoHS 2011/65/EU

### Harmonised Standards

| Area              | Standards                                          |
|-------------------|----------------------------------------------------|
| EMC               | EN 61000-6-1 (Immunity) · EN 61000-6-3 (Emissions) |
| Electrical Safety | EN 62368-1                                         |
| Radio             | EN 300 328 · EN 301 489-1 · EN 301 489-17          |
| RoHS              | EN IEC 63000                                       |

### Radio

The product integrates a pre-certified ESP32 Wi-Fi radio module (2.4 GHz). Final product conformity with the Radio Equipment Directive is demonstrated by the maintained technical documentation and conformity assessment of the complete device.

### Safety notice

L / N: hazardous voltage · 24 V DC: SELV · Qualified personnel only

### Published by

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